

Bluetooth Audio Adaptor — Design Overview

Rev B (in design) · Updated 2026-08-28 · Jason Describes the **target design**. Deltas from as-built Rev A are listed in §10.

Living document. Every confirmed design change lands here. Planning, open questions and review findings live in 0 and 1 — not here.

1. Function

Bluetooth audio adaptor for a car AUX input. Powered from USB-C or an internal Li-ion cell.

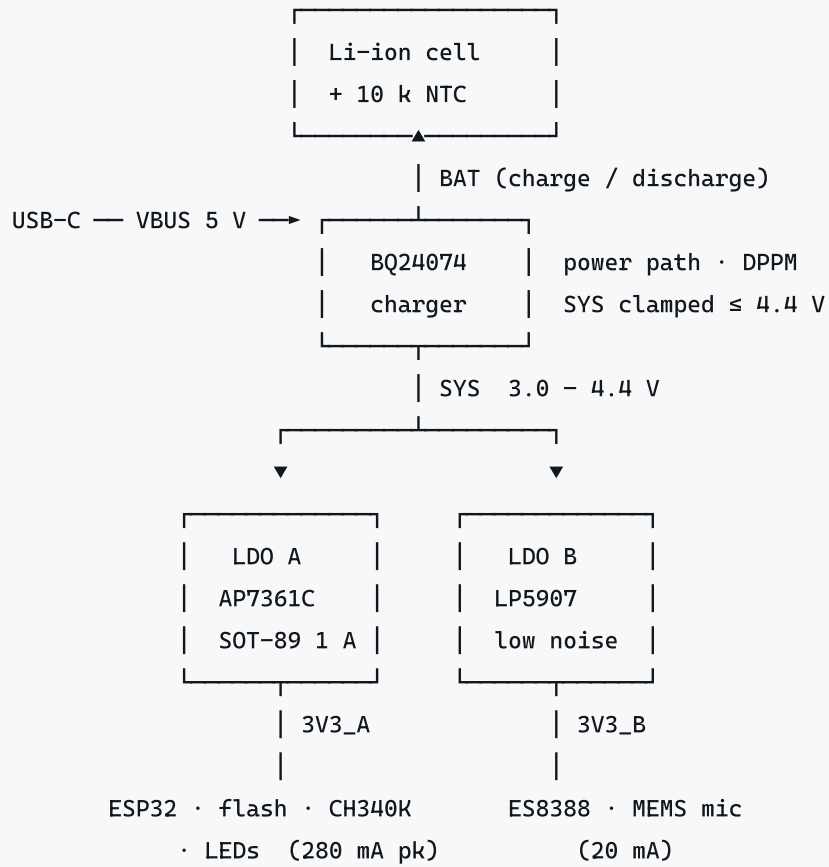
Mode	Path	Status
RX (primary)	Phone → Bluetooth A2DP → codec DAC → 3.5 mm out	Rev A hardware complete
TX	3.5 mm in → codec ADC → Bluetooth → headphones/speaker	Rev B addition
HFP handsfree	On-board mic → codec ADC → BT; BT → DAC → jack	Rev B addition

Key constraint: must work with all phones, not just recent ones → **Bluetooth Classic (A2DP/HFP), not BLE Audio**. This single requirement determines the MCU (\$5).

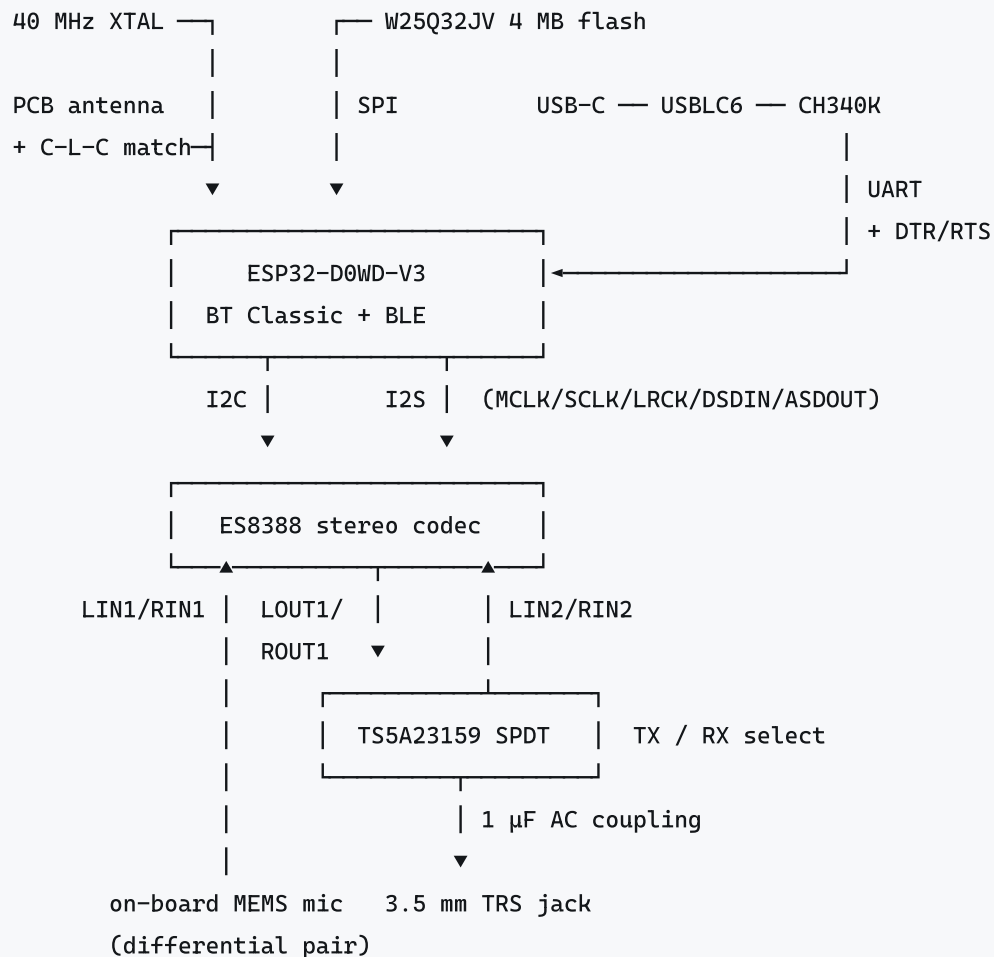
Out of scope: amplified speaker output, multipoint pairing, app control.

2. Architecture

Power tree



Signal chain



Rails

Rail	Source	Loads	Why separate
SYS 3.0–4.4 V	BQ24074	Both LDOs	Power path: USB when present, battery when not
3V3_A	LDO A	ESP32, flash, CH340K, LEDs	Carries RF current bursts
3V3_B	LDO B (LP5907)	ES8388, mic	Isolates a 96 dB codec from those bursts

3. Power subsystem

Linear regulation throughout. A switching converter was rejected: this is a mixed-signal audio board and the efficiency gain does not pay for the noise.

Part	Function	Key spec
BQ24074RGTR (C54313)	Power-path charger	1.5 A, SYS clamped ≤ 4.4 V, DPPM, NTC input, 10.5 V OVP
AP7361C-33Y5-13 (C460397)	LDO A — digital/RF	SOT-89-5, 1 A, ultra-low dropout
LP5907MFX-3.3 (C80670)	LDO B — codec analog	6.5 μV_{RMS} , high PSRR, no bypass cap needed
USBLC6-2SC6	USB ESD	In-line on D+/D–
Li-ion cell + 10 k NTC	Energy	Removable. NTC bonded to cell, not PCB

Charger behaviour. System load takes priority over charge current (DPPM). SYS follows USB when present, battery when not. The $\leq 4.4\text{ V}$ clamp is what makes LDO A's thermals work (§7).

Safety. Charging is inhibited outside the cell's temperature window in hardware via the NTC on **TS** — not in firmware. Cell is removable; device is not stored in the vehicle.

Why not LFP: 3.2 V nominal sits below the 3.3 V rail for most of its discharge, forcing a boost converter in front of the codec.

4. Audio subsystem

ES8388 stereo codec — 24-bit, 96 dB DAC dynamic range, -83 dB THD+N . Chosen over the ESP32's internal DAC for quality, and over a DAC-only part because HFP needs an ADC.

Signal	Connection
Control	I ² C, address 0x20 (CE pulled to DGND)
Audio	I ² S, codec as slave ; full duplex for HFP
Clocks	MCLK, SCLK, LRCK from ESP32
Output	LOUT1/ROUT1 $\rightarrow 1\text{ }\mu\text{F} \rightarrow 10\text{ }\Omega \rightarrow$ analog switch $\rightarrow$ jack
Input (TX)	Jack $\rightarrow$ switch $\rightarrow$ LIN2/RIN2
Input (mic)	Differential LIN1/RIN1

All four codec rails run at 3.3 V. DVDD/PVDD at 1.8 V would put the codec's I/O in a different logic domain from the 3.3 V ESP32 — see §7.

Part	Function	Key spec
ES8388	Stereo codec	QFN-28, 0.4 mm; AVDD/HPVDD/DVDD/PVDD all 3V3
SPH8878LR5H-1 (C3171733)	MEMS microphone	LGA-6, differential or single-ended, MSL 1
TS5A23159DGSR (C42751)	TX/RX path switch	Dual SPDT, $1\text{ }\Omega$ R_{on} , break-before-make
3.5 mm TRS jack	Audio I/O	3-pole

Output coupling. $1\text{ }\mu\text{F}$ blocks the VMID bias. Into a $\sim 10\text{ k}\Omega$ AUX load that is a $\sim 16\text{ Hz}$ corner. Use a case size large enough that Class-II dielectric distortion stays below the codec's own THD.

Ground loop. Chassis ground (USB charger) and head-unit audio ground form a loop $\rightarrow$ alternator whine. Mitigations: run on battery with USB unplugged (free), or fit 1:1 isolation transformers in place of R7/R8. Both provisioned; footprints accept either.

5. Microcontroller & RF

ESP32-D0WD-V3. QFN-48, 0.35 mm pitch, dual-core Xtensa LX6.

Selected for firmware capability, not silicon features: it is the only Espressif part whose radio hardware supports **Bluetooth Classic**, and the only one whose SDK provides an **A2DP sink** and **HFP**. ESP32-S3/C3/C6/H2 and the entire Nordic nRF range are BLE-only. This choice follows directly from §1's compatibility requirement.

Part	Function	Key spec
ESP32-D0WD-V3	MCU + radio	BT Classic + BLE + Wi-Fi; needs external flash
W25Q32JV	Program store	4 MB, 3.3 V (JV not JW), on VDD_SDIO
40 MHz crystal	Reference	±10 ppm required; load caps per §7
PCB inverted-F antenna + C-L-C match	2.4 GHz	50 Ω controlled impedance, 4-layer

RF front end is a deliberate learning exercise. A pre-certified module would remove the matching network, crystal trim and impedance routing — that is precisely the work being kept. Requires a NanoVNA (antenna S11) and a frequency-offset measurement to close.

Stackup: 4-layer. Signal / GND / PWR / Signal, with a solid ground plane under the RF section.

6. Programming & debug

No USB peripheral on this ESP32 variant, and no SWD (Xtensa, not ARM).

Path	Parts	Use
USB auto-program	CH340K + UMH3N dual transistor on DTR/RTS	Day-to-day flashing
Off-board serial	J3 header (DNP)	Fallback
JTAG	J4 2×5 header (DNP), ESP-PROG	Debugging

Headers are DNP — fit before a debug session. 13 test points cover I²C, mic, audio out, boot, GPIO and rails.

7. Key calculations

Power budget

Load	Rail	Current
ESP32, BT Classic streaming (avg)	A	130 mA
ESP32 transmit peak	A	250 mA
SPI flash	A	5 mA
CH340K (USB attached only)	A	10 mA
2× LED @ 330 Ω	A	8 mA
Rail A		155 mA avg · 280 mA peak
ES8388 @ 3.3 V, playback + record	B	18 mA
MEMS mic	B	0.4 mA
Rail B		~20 mA

LDO A thermal. Bluetooth bursts last 1–2 ms; package time constant is seconds, so the junction integrates to the average, not the peak.

$$P_{\text{avg}} = (4.4 - 3.3) \times 0.155 = 0.17 \text{ W}$$

SOT-89, $\theta_{JA} \approx 100 \text{ }^{\circ}\text{C/W}$ $\rightarrow \Delta T = 17 \text{ }^{\circ}\text{C} \rightarrow T_j \approx 87 \text{ }^{\circ}\text{C}$ at $70 \text{ }^{\circ}\text{C}$ ambient ✓

SOT-23-5, $\theta_{JA} \approx 200 \text{ }^{\circ}\text{C/W}$ $\rightarrow \Delta T = 34 \text{ }^{\circ}\text{C} \rightarrow T_j \approx 104 \text{ }^{\circ}\text{C}$ ✓ but tight

LDO A dropout $\rightarrow$ usable battery. Dropout, not current, is the binding spec.

$$\text{Cutoff} = 3.3 \text{ V} + V_{\text{dropout}}(280 \text{ mA})$$

150 mV $\rightarrow 3.45 \text{ V} \rightarrow \sim 87 \%$ of cell

250 mV $\rightarrow 3.55 \text{ V} \rightarrow \sim 83 \%$

400 mV $\rightarrow 3.70 \text{ V} \rightarrow \sim 75 \%$

1.2 V $\rightarrow 4.50 \text{ V} \rightarrow$ will not run on battery (NCP1117, Rev A – removed)

Runtime (LDO path, $\sim 83 \%$ usable): $1000 \text{ mAh} \approx 4.8 \text{ h} \cdot 2000 \text{ mAh} \approx 9.7 \text{ h} \cdot 3000 \text{ mAh} \approx 14.5 \text{ h}$.

Crystal load capacitance.

$$CL = (C1 \times C2)/(C1 + C2) + C_{\text{stray}} \quad C_{\text{stray}} \approx 2\text{--}5 \text{ pF}$$

Rev A: $18 \text{ pF} \parallel 18 \text{ pF} \rightarrow 9 + 3 \approx 12 \text{ pF}$ against a 10 pF crystal $\rightarrow$ over-loaded, frequency low

Rev B: $\approx 15 \text{ pF} \parallel 15 \text{ pF} \rightarrow 7.5 + 3 \approx 10.5 \text{ pF}$, then trim on measured offset ($\pm 10 \text{ ppm}$ required)

Logic-level compatibility (why all codec rails are 3.3 V).

$$\text{ESP32} \quad V_{IH}(\text{min}) = 0.75 \times 3.3 = 2.475 \text{ V} \quad V_{IL}(\text{max}) = 0.825 \text{ V}$$

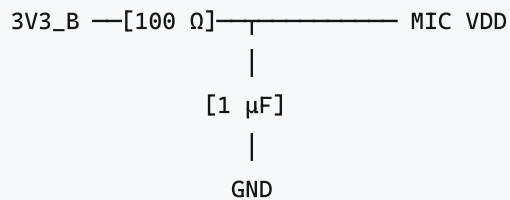
Codec at PVDD 1.8 V: $V_{OH} \approx 1.8 \text{ V} \rightarrow$ lands between V_{IL} and V_{IH} = undefined

Codec abs-max input = $DVDD + 0.3 \text{ V} = 2.1 \text{ V} \rightarrow 3.3 \text{ V}$ drive exceeds it

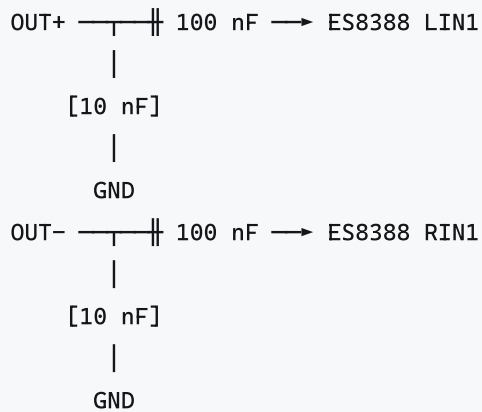
Audio output coupling. $f = 1/(2\pi \cdot 10 \text{ k}\Omega \cdot 1 \text{ }\mu\text{F}) \approx 16 \text{ Hz}$.

8. Conceptual circuits

Microphone — differential, no external bias. MEMS outputs are internally biased ($0.66/0.70 \text{ V}$, $340/410 \text{ }\Omega$). An electret-style bias resistor fights the internal amplifier and costs headroom.



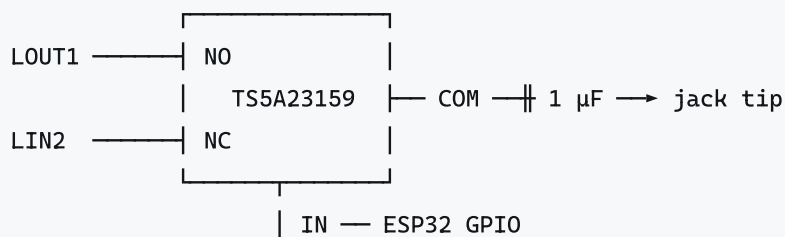
RC filter: mic PSRR is only -45 dB



Route OUT+/OUT- as a matched parallel pair - common-mode rejection depends on both picking up identical noise.

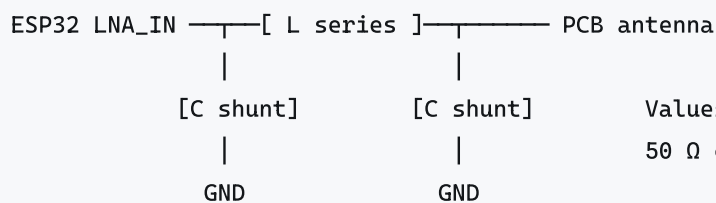
$10\text{ nF} + 340\text{ }\Omega \approx 47\text{ kHz}$ - RF shunt only, above the audio band.

TX/RX switch — placement is critical.



Switch sits on the **codec side** of the coupling capacitor. Both throws are at VMID (~1.65 V), inside the switch's 0–3.3 V supply rails. On the jack side the signal is centred at 0 V and every negative half-cycle would clip.

RF match — C-L-C Pi, per Espressif.



Values tuned on measured S11.
50 Ω controlled-impedance trace.

9. Firmware dependencies

Hardware choices that constrain firmware, or vice versa.

Item	Note
Bluetooth Classic	Only the original ESP32 has Classic radio hardware. A2DP sink + HFP come from ESP-IDF/ESP-ADF. Drives the entire MCU selection.
Pairing persistence	Bluedroid writes bonding keys to NVS in flash automatically. Survives power-off with no battery. Requires an initialised NVS partition.
MCLK pin	ESP32 can emit MCLK only on GPIO0/1/3. GPIO1/3 are the UART, so GPIO0 — which is also the BOOT strapping pin. Safe: the codec's MCLK input is high-Z at reset.
ADC	Use ADC1 only (battery sense). ADC2 is unusable while the radio is active. GPIO32–39 free.
Flash mode	Configure DIO . QIO requires IO2/IO3 correct — fixed in Rev B, but DIO is the safe default.
JTAG + GPIO12	MTDI is the VDD_SDIO strapping pin. A debugger driving TDI high across reset selects 1.8 V and the 3.3 V flash fails. Fix: burn <code>XPD_SDIO_*</code> eFuses to force 3.3 V. Irreversible .
Codec I²C address	0x20 (<code>CE</code> low). ESP-ADF's ES8388 driver default.
I²S full duplex	Required for HFP. Configure both TX and RX channels.
TX/RX switch	One GPIO selects the audio path. Firmware must track mode.
Deep sleep	If used for battery standby, populate the CAP2 RC network.

10. Deltas from Rev A (as built)

#	Change	Reason
1	Codec rails 1.8 V → 3.3 V; delete 1.8 V LDO, add dedicated codec LDO	Logic-level incompatibility + supply noise
2	<code>CE</code> pulled to DGND	Was floating — indeterminate I ² C address
3	Add 10 nF on ESP32 <code>CAP1</code>	Required by datasheet; was absent
4	Swap flash IO2/IO3	Crossed; corrupts QIO reads
5	Delete mic bias network (R21/R22/R23/C43/C44); add RC on mic VDD	Electret circuit on a self-biased part
6	Mic → SPH8878LR5H-1	ICS-40720 discontinued
7	LED resistors 1 kΩ → 330 Ω	~0.3 mA was invisible
8	TRRS → 3-pole TRS jack	4th pole unused
9	Add TX input path + analog switch	TX mode now in scope
10	Add battery charger, NTC, cell connector, VBAT sense	Battery in scope
11	NCP1117 → low-dropout LDO in SOT-89	1.2 V dropout cannot run from a cell
12	Crystal load caps 18 pF → ~15 pF, then trim	Over-loaded; ±10 ppm required

Still open: ground-loop mitigation (transformers vs battery-only playback); crystal offset measurement; RF match tuning; BOM line consolidation.

11. Manufacturing

JLCPCB, 4-layer, **Economic PCBA** with the ESP32 excluded from assembly and hand-populated (it is the only Standard-Only part). All other parts verified Economic-tier.

Constraints: single-sided placement (satisfied — nothing on the back is assembled), 0402 minimum passive, 0.4 mm minimum IC pitch.

Build: 5 assembled + 5 bare. Target \$20 USD landed per unit.

Hand assembly: 0.35 mm-pitch QFN requires a laser-cut stencil, plugged/capped via-in-pad on the thermal pad (open vias wick solder and degrade the RF ground), and ~50–70 % paste coverage on the exposed pad.

12. References

Local ([hw/datasheets/](#), [hw/app notes/](#))

- [ESP32 datasheet](#) · [ESP32 Hardware Design Guidelines](#) · [ESP32 Technical Reference Manual](#)
- [ES8388 datasheet rev 5.0](#) · [ES8388 User Guide](#)
- [W25Q32JV datasheet rev G](#)
- [ESP32-LyraT v4.3 schematic](#) — reference design for the codec section
- [AN043 2.4 GHz PCB Antenna](#) · [Antennas for IoT](#)

External

- [JLCPCB PCB assembly capabilities](#)
- [ESP-ADF ES8388 driver](#)
- [esptool — set flash voltage eFuses](#)

Project

- [0](#) — requirements, constraints, traceability
- [0](#) — netlist audit and findings
- [0](#) — battery architecture
- [0](#) — mic theory and selection

13. Revision history

Rev	Date	Change
A	2025-04-05	As-built schematic, commit 369c5f9
B	2026-08-24	This document created. Scope decisions resolved; 12 deltas in §10